

We are continuously innovating in terms of technology to reduce the consumption of power in silicon and new materials like SiC

sume less energy. And then there is the trend to exploit fusion of alternative energies. Take solar energy, for instance. We have developed a micro inverter for solar cell solutions that is particularly intended to optimise the power efficiency of solar cells. In this frame, we have developed a single-chip solution that is kind of power tracking solution to really optimise the efficiency at the level of one cell, excluding areas of the cell that aren't well exposed to the sun, for instance, and boosting of the rest.

There are many things that you can do; for instance, in automobiles, to start and stop, or vice versa in the city. It happened to me when I was in the gulf a few days ago. A car automatically stopped on approaching the traffic light. This is another example of fuel consumption reduction.

Coming to ST, how much you value innovation while designing?

We have taken many initiatives at ST to boost innovation. For instance, we have a specific organisation that we call Advanced System Technology that works on future trends just to explore new opportunities and new ideas for innovation. Every year, we meet with the council of well-known professors from universities to benchmark what we do. We have innovation circles in many sites where we operate to boost innovation. We have a corporate initiative called cross-functional initiative that we define as R&D excellence. This cross-functioning initiative again is to boost the effectiveness of what we do in R&D and therefore to boost innovation. The objective is to be more innovative and drive radical innovation.

How important is it for you to have design centre in India?

The quantity of what we do here is as important as the quality of what we do here. Here the market is becoming very significant and therefore we need to be present developing products for the Indian market also.

We have 2200 people in our Greater Noida and Bengaluru centres. And I think we are one of the biggest in the semiconductor arena in India. We are working on many important programmes not only for digital consumer and computer peripheral applications but also for automobiles where we are working on advanced 32nm technology line powertrain. We are working on powertrain, entertainment and navigation chips. ST India contributes 15 to 20 per cent of the overall design portfolio of the company.

What are the three key areas of growth here?

The three key areas of growth are digital consumers

(combination of set-top box and digital TV), power (lighting and electronic metering), and security (identification and trust using secure microcontrollers, smart cards, ID cards, etc).

What is your strategy to win the market share?

The strategy is to focus on the four application areas that I described earlier. Today, I think we are No. 3 and by focusing on these areas, we can become No. 1.

We have five product lines: Platforms for cellphones, ASICs, microcontrollers, analogue and MEMS, and also power products.

Now that there is greater focus on green design, what green solutions do you have?

Everything that we do in the area of alternative energy is very, very green. Not only is it green, but we are contributing to make the green 'greener'. The solar cell is already green, but we have developed our solutions to make the solar cell more efficient.

What advantages does ST enjoy over other semiconductor companies in India?

Well, first of all, our range of technologies and products is very broad. We serve all the semiconductor markets, with two exceptions—one is flash memory and DRAM and the second is the microprocessor for the PC motherboard. In India, there is no real business on PC motherboard. So all that we do fits very well with the requirements of our Indian customers.

With the vast portfolio of our products, we can address a number of applications that are expanding very rapidly. Also, the 2200-strong staff is a massive presence that we have developed year after year. There are very few companies with this size of investment and R&D in India.

Your key priorities this year?

First, we will continue to expand and grow our customer base, with additional focus on Asia overall.

Second, a focused but significant capital investment in capacity for MEMS, in smart applications for the automotive field and in the U8500 smartphone platforms.

Third, after the completion of their restructuring, 2011 will be the year of ST-Ericsson's transition, when the transformed, new product portfolio will start ramping from the second half of this year.

Fourth, we want to achieve breakthroughs with a number of new, innovative products, including system-on-chips for 3D and connected TVs; microcontrollers based on the Power PC architecture for automotive applications; twin MEMS (combining gyroscopes and accelerometers either in a package or with innovative solutions); new MEMS families like active microphones and pressure sensors; advanced analogue products; and the U8500 smartphone, tablet and thin modem platforms coming from ST-Ericsson. ●